



No more jobs at Twitter

The company reportedly laid off 3,738 people out of its total workforce of 7,500. <https://digit.in/nov22-29>



Insta NFTs

Instagram will soon allow users to mint and sell NFTs. The concept is being tested by the developers. <https://digit.in/nov22-30>

Intel Core i9-13900K

It's neck and neck again

Intel 13th Gen Raptor Lake desktop processors have their work cut out for themselves.

The 13th Gen processors have to outdo the AMD Zen 4 Raphael processors and that's no easy task. 13th Gen Intel processors feature the same hybrid microarchitecture that the 12th Gen processors have but with a lot more E-Cores than before. So, were they able to do so? Let's find out in this review.

SPECIFICATIONS

Just like the previous gen, even the 13th Gen launch lineup have six processors at launch. There are two SKUs under each performance tier – Core i9, Core i7 and Core i5. The Core i3 processors might launch later or may not. The 12th Gen Core i3 processors launched after a significant amount of time after the initial launch period. This is quite common even with AMD's budget tiers. The six processors launched are the K and KF processors. The integrated GPU is the same Intel UHD 770 graphics but the clock speeds have been bumped up by 100 MHz. So there will be a little bit of extra gaming performance for folks relying on the iGPU. The bulk of the change with the 13th Gen Intel Raptor Lake processors is the doubling of the E-Cores. The 13th Gen processors are being shipped with 16 E-Cores. The total thread count of the flagship Intel Core i9-13900K becomes 32 considering that all the P-Cores have hyperthreading whereas the E-Cores don't.

PERFORMANCE

The Core i9-12900K has a max turbo power of 241 watts and motherboard manufacturers tend to ignore such limits to boost performance. As always, we begin with Cinebench R23 to see how the processor fares in single threaded and multithreaded performance. In the single-threaded benchmark, the Intel Core i9-13900K scores 2264 which is a decent gain over



the newly released AMD Ryzen 9 7950X which scores 1997. The Intel Core i9-12900K had scored 1971. However, in the multi-threaded run we see the AMD processor retains the lead with just 40 points. V-Ray is a 3D modelling software by Chaos that can focus on just the CPU and the GPU separately. Yet again, the Intel Core i9-13900K certainly beats the 12900K but the AMD Ryzen 9 7950X continues to lead with a score of 29280. The 13900K scored 25332 and the 12900K had scored 17649. Blender has standard benchmarking scenes and we use multiple iterations of the Classroom, Monster and Junkshop scenes for our tests. Here, the AMD Ryzen 9 7950X takes the lead yet again. The Intel Core i9-13900K does outperform the Intel Core i9-12900K by a significant margin. 7-Zip is an open-source archiving software with really good and frequently updated algorithms that perform compression and decompression. It also scales well with higher core-counts so there's an added advantage with 7-Zip over other archiving software. It shows the Intel Core i9-13900K gaining a significant lead over the 12900K but is yet again bested by the Ryzen 9 7950X.

Mozilla Kraken is a browser-based JavaScript benchmark that executes several real-world scenarios that everyone comes across while normally browsing the Internet. Intel usually excels at Mozilla Kraken and even Octane Javascript benchmark. Intel continues to maintain its lead across both benchmarks over the AMD processor. Office productivity hinges upon some popular office applications along with browsing. Microsoft's Office suite is the most popular of them all and the benchmark by UL is focused purely on that. Here, the Intel Core i9-13900K is a clear winner beating its predecessor and the Zen 4 flagship with a decent margin. Gaming is where

Intel has traditionally held the top spot. In our benchmarks we see the 13900K and the 7950X be neck and neck.



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VERDICT

The Intel Core i9-13900K has done something very significant in outperforming the 12th Gen 12900K by a significant margin within just one

generation. However, across a lot of benchmarks, the AMD Ryzen 9 7950X maintains a lead over the 13900K. In a lot of benchmarks, the difference isn't much. So for enthusiasts, there are now two very competitive high-end desktop processors to choose from. That's a good place to be in.

—Mithun Mohandas

SPECIFICATIONS

TOTAL CORES: 24 | # OF PERFORMANCE-CORES: 8 | # OF EFFICIENT-CORES: 16 | TOTAL THREADS: 32 | MAX TURBO FREQUENCY: 5.80 GHz | CACHE: 36 MB Intel® Smart Cache | TOTAL L2 CACHE: 32 MB | PROCESSOR BASE POWER: 125 W | MAXIMUM TURBO POWER: 253 W | MAX MEMORY SIZE (DEPENDENT ON MEMORY TYPE): 128 GB | MEMORY TYPES: Up to DDR5 5600 MT/s, Up to DDR4 3200 MT/s | MAX # OF MEMORY CHANNELS: 2 | PROCESSOR GRAPHICS: Intel® UHD Graphics 770 | GRAPHICS BASE FREQUENCY: 300 MHz | GRAPHICS MAX DYNAMIC FREQUENCY: 1.65 GHz | GRAPHICS OUTPUT: eDP 1.4b, DP 1.4a, HDMI 2.1 | EXECUTION UNITS: 32 | MAX RESOLUTION (HDMI): 4096 x 2160 @ 60Hz | MAX RESOLUTION (DP): 7680 x 4320 @ 60Hz | SOCKETS SUPPORTED: FCLGA1700 | MAX CPU CONFIGURATION: 1

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